

The μ OS++ test catalogue

Every test, per architecture, platform and board

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Every test the harness registers, per architecture, platform and board: what each test does, whether it runs the SMP scheduler or a single core, how it is run (QEMU, host or the real board) and, for the board runs, which debug probe.

The sources of truth are the code, not other documents:

What	Where
which cases exist	<code>tests/platforms/<platform>/CMakeLists.txt</code> ; for port platforms, the directories under <code><port>/test/<board>/</code> (globbed at configure time)
per-test knobs (core count, HWD_ONLY, sources)	<code><port>/test/<board>/tests.cmake</code>
board facts (CPU, core count, QEMU linker)	<code><port>/test/boards/<board>/board.cmake</code>
probe	<code><port>/test/boards/<board>/openocd*.cfg</code> and <code>hw.sh</code> ; for the upstream platforms, the <code>openocd</code> command in <code>tests/platforms/<platform>/CMakeLists.txt</code>
xpm actions	<code>tests/package.json</code>

The ports are the sibling repositories `micro-os-plus-iii-aarch32`, `-aarch64`, `-cortexm`, `-devices` and `-posix-arch` (checked with or without `.git` suffix); the unified SMP kernel and harness root is `micro-os-plus-iii` (branch `smp`), with tests driven from `package.json` at root or under `tests/`.

1. Legend

Variant (the suffix of the CTest name):

Suffix	Runs on	Runner
<code>-qemu</code>	QEMU	<code>test_smpl/run-qemu.sh</code> , or QEMU directly (Cortex-M)
<code>-host</code>	the Linux host (POSIX port)	<code>test_smpl/run-host.sh</code> , or the executable directly
<code>-hwd</code>	the real board, through OpenOCD	the board's <code>hw.sh</code> (then <code>test_smpl/run-hw.sh</code> on the Pis)
<code>-test</code> (no variant)	harness-only platforms: QEMU, or the board through OpenOCD	the command in the platform's <code>CMakeLists.txt</code>

`-hwd` cases carry the `hwd` label and are excluded from `xpm run test` (`ctest -LE hwd`). They are run one at a time, one per power cycle, with OpenOCD only (no GDB); the verdict is the `RESULT: PASS|FAIL` line, read from semihosting and mirrored on the UART.

Scheduling (the "Mode" column below):

Mode	Meaning
SMP n	kernel built with <code>OS_USE_SMP_SCHEDULER</code> , <code>OS_NCPU = n</code> , all n cores scheduling
SMP n, 1 active	SMP build, but the test keeps the secondaries parked
SMP x1	SMP scheduler code paths at <code>OS_NCPU = 1</code> (single-CPU board)
single	the kernel's non-SMP branch (<code>OS_USE_SMP_SCHEDULER</code> undefined), one core
bare	no kernel at all: a bare-metal probe

2. Platforms at a glance

Platform (xpm config prefix)	Port	Board / machine	CPU	Cores	Variants	Probe
<code>aarch32-rpi-zero-2w</code>	aarch32	Raspberry Pi Zero 2 W (BCM2837)	4× Cortex-A53, AArch32	4	qemu, hwd	J-Link (default) or Olimex
<code>aarch32-rpi3b</code>	aarch32	Raspberry Pi 3 B (BCM2837)	4× Cortex-A53, AArch32	4	qemu, hwd	J-Link (default) or Olimex
<code>aarch32-luckfox-lyra</code>	aarch32	Luckfox Lyra B (RK3506)	3× Cortex-A7 (+ a Cortex-M0 not scheduled)	3	hwd only	WCH-Link, CMSIS-DAP
<code>aarch64-rpi-zero-2w</code>	aarch64	Raspberry Pi Zero 2 W (BCM2837)	4× Cortex-A53, ARMv8-A	4	qemu, hwd	J-Link (default) or Olimex
<code>aarch64-rpi3b</code>	aarch64	Raspberry Pi 3 B (BCM2837)	4× Cortex-A53, ARMv8-A	4	qemu, hwd	J-Link (default) or Olimex
<code>cortexm-pico2</code>	cortexm	Raspberry Pi Pico 2 (RP2350)	2× Cortex-M33	2	qemu (6 tests), hwd, ram (4 tests)	any CMSIS-DAP
<code>cortexm-pico2-pizero</code>	cortexm	Pi-Zero RP2350B, 16 MB flash	2× Cortex-M33	2	hwd only	XV-Link CMSIS-DAP
<code>cortexm-pico2-rp2350b-psram</code>	cortexm	WeAct RP2350B, 16 MB flash + 8 MB PSRAM	2× Cortex-M33	2	qemu (2 tests), hwd	CMSIS-DAP <code>c251:f001</code>
<code>cortexm-nucleof411</code>	cortexm	ST Nucleo-F411RE	1× Cortex-M4F	1	qemu (netduinoplus2, 4 tests), hwd	on-board ST-Link v2.1
<code>cortexm-weactf411</code>	cortexm	WeAct Studio F411CE	1× Cortex-M4F	1	qemu (netduinoplus2, 5 tests), hwd	DAPLink CMSIS-DAP (or WCH-Link)
<code>cortexm-weactf412</code>	cortexm	WeAct Studio F412RE	1× Cortex-M4F	1	hwd only (SRAM 256K > QEMU 128K)	ST-Link
<code>native</code>	posix-arch	the Linux host	host threads as CPUs	4 (<code>-DNCPU</code>)	host	—
<code>2xcortex-m33</code>	cortexm	QEMU <code>mps2-an521</code>	2× Cortex-M33	2	QEMU	—
<code>pico2-1cpu</code>	cortexm	QEMU <code>mps2-an505</code>	1× Cortex-M33	1	QEMU	—
<code>qemu-cortex-m0</code>	cortexm	QEMU <code>mps2-an385</code>	Cortex-M3 (the platform passes <code>--cpu</code>)	1	QEMU	—

Platform (xpm config prefix)	Port	Board / machine	CPU	Cores	Variants	Probe
			cortex-m3 , not an M0)			
qemu-cortex-m3	cortexm	QEMU mps2-an385	Cortex-M3	1	QEMU	—
qemu-cortex-m4f	cortexm	QEMU mps2-an386	Cortex-M4	1	QEMU	—
qemu-cortex-m7f	cortexm	QEMU mps2-an500	Cortex-M7	1	QEMU	—
nucleo-f411re	harness (upstream)	ST Nucleo-F411RE	Cortex-M4F	1	board	ST-Link (interface/stlink-dap.cfg)
nucleo-f767zi	harness (upstream)	ST Nucleo-F767ZI	Cortex-M7	1	board	ST-Link (interface/stlink-dap.cfg)
nucleo-h743zi	harness (upstream)	ST Nucleo-H743ZI	Cortex-M7	1	board	ST-Link (interface/stlink-dap.cfg)
raspberrypi-pico	harness (upstream)	Raspberry Pi Pico (RP2040)	Cortex-M0+	1	board	CMSIS-DAP (interface/cmsis-dap.cfg)

Every platform has a `-cmake-gcc-debug` and a `-cmake-gcc-release` configuration; `native` also has `-cmake-sys-*`, `-cmake-gccNN-*` and `-cmake-clangNN-*`. Only these configurations carry per-test `test-<app>-<variant>` actions: the `cortexm-*`, `aarch32-*`, `aarch64-*`, `qemu-cortex-*`, `2xcortex-m33`, `pico2-1cpu`, `native-cmake-gcc-*` and `native-cmake-sys-*` ones (the release configurations inherit them from debug).

The QEMU machine for the Pis is `raspi3b -smp 4`; the AArch32 images are started through a small AArch64 boot shim (`shim8.img`, image at `0x10000`). The Cortex-M `-qemu` images of `cortexm-pico2` and `cortexm-pico2-rp2350b-psram` run on `mps2-an500 -cpu cortex-m7`.

The four `qemu-cortex-*` platforms link the local Cortex-M port's generic single-core QEMU cores (`micro-os-plus::cortexm-qemu-m0/-m3/-m4f/-m7`) and the generic device in `tests/device-qemu-cortexm`, and run the SMP kernel single-core (`OS_NCPU=1`, no `OS_USE_SMP_SCHEDULER`). Each registers three suites: `<platform>-rtos-apis-test`, `-mutex-stress-test` and `-cmsis-os-validator-test`. (`2xcortex-m33` and `pico2-1cpu` likewise link the local generic M33 core, `micro-os-plus::cortexm-qemu-m33`.)

3. Probes

Board	Probe	USB id	Transport, speed	Config
Pi Zero 2 W, Pi 3 B (aarch32 and aarch64)	SEGGER J-Link — default	—	JTAG	<code>test/boards/rpi-zero-2w/openocd-jlink-rpi3.cfg</code>
Pi Zero 2 W, Pi 3 B, with <code>PROBE=olimex</code>	Olimex ARM-USB-OCD (FTDI)	—	JTAG, 1000 kHz	<code>test/boards/rpi-zero-2w/openocd-olimex.cfg</code>
Luckfox Lyra B	WCH-Link in ARM mode (CMSIS-DAP)	<code>1a86:8011</code>	SWD, 4000 kHz	<code>test/boards/luckfox-lyra/openocd.cfg</code>
Pico 2	any CMSIS-DAP (auto-selected, no vid_pid pinned)	—	SWD, 1000 kHz	<code>test/boards/pico2/openocd.cfg</code>
Pi-Zero RP2350B	XV-Link CMSIS-DAP	<code>0416:5951</code>	SWD, 2000 kHz	<code>test/boards/pico2-pizero/openocd.cfg</code>
WeAct RP2350B + PSRAM	CMSIS-DAP	<code>c251:f001</code>	SWD, 1000 kHz	<code>test/boards/pico2-rp2350b-psram/openocd.cfg</code>
Nucleo-F411RE	on-board ST-Link v2.1	—	SWD, <code>connect_assert_srst</code>	<code>test/boards/nucleof411/openocd.cfg</code>
WeAct F411CE	DAPLink CMSIS-DAP (override: WCH-Link <code>1a86:8011</code> through <code>DAP_VID_PID</code>)	<code>0d28:0204</code>	SWD, 1000 kHz (<code>ADAPTER_KHZ</code>)	<code>test/boards/weactf411/openocd.cfg</code>
WeAct F412RE	ST-Link V2 / V2-1 / V3	by class	SWD (<code>ADAPTER_KHZ</code>)	<code>test/boards/weactf412/openocd.cfg</code>

The Pi 3 B `hw.sh` is a wrapper that `exec`s the Zero 2 W's, so both Pis take the same `PROBE` switch. The paths above are relative to the port repository (`aarch32` or `aarch64` for the Pis and the Lyra, `cortexm` for the rest).

4. The harness suites (portable)

Their sources live in `tests/sources/<suite>/`. They have no `main()`: the entry is `os_main()`, and its return code is the verdict. On the port boards a per-board wrapper wraps `os_main` (`-Wl,--wrap=os_main`), prints the `RESULT:` line and stops the run. It is each suite's `harness-suite.cpp` on the Pi ports, and one `test/<board>/harness-suite.hpp` per board, included by the suites' `harness-suite.cpp`, on cortexm.

Suite	What it does
<code>rtos-apis</code>	Exercises the μ OS++ C++ API, C API, ISO (<code>std::</code>) API, CMSIS-RTOS v1 wrapper and the POSIX I/O layer, plus a Chan FatFs test.
<code>mutex-stress</code>	Several threads take one mutex at random intervals; the test checks that every thread got it and the distribution is sane.
<code>fp-switch</code>	Six unpinned threads at three priorities each load their own pattern into s0-s31 and the FPSCR flags, spin long enough to be preempted (and, on SMP, moved to the other core), and check the registers read back unchanged. Needs an FPU; without one it passes with nothing to test.
<code>cmsis-os-validator</code>	The Arm CMSIS-RTOS v1 validator (60 test cases: threads, timers, signals, semaphores, mutexes, memory pools, message and mail queues, with interrupt-context cases). On a Cortex-M the interrupt is NVIC IRQ 0; see below for the other cores.
<code>blinky</code>	Blinks the LED. Enabled only on <code>nucleo-f411re</code> (<code>platforms/nucleo-f411re/cmake/definitions.cmake</code>), where it is the case <code>blinky-test</code> .
<code>instrumentation</code>	SEGGER SystemView use cases. Enabled only on <code>nucleo-f411re</code> , which builds <code>instrumentation-test</code> and <code>rtos-apis-instrumentation-test</code> but registers no CTest case for them: they are run from SEGGER Ozone with a J-Link (see <code>tests/sources/instrumentation/README.md</code>).

Where each suite runs:

Platform	rtos-apis	mutex-stress	cmsis-os-validator	Mode
aarch32-rpi-zero-2w, aarch32-rpi3b, aarch64-rpi-zero-2w, aarch64-rpi3b	qemu, hwd	qemu, hwd	qemu, hwd	SMP 4
aarch32-luckfox-lyra	—	hwd (mutex-stress-test)	—	SMP 3
cortexm-pico2	qemu, hwd, ram	qemu, hwd, ram	qemu, hwd, ram	SMP 2 on hwd; single on qemu & ram (SRAM load)
cortexm-nucleof411, cortexm-weactf411	qemu, hwd	qemu, hwd	qemu, hwd	SMP ×1
cortexm-weactf412	hwd	hwd	hwd	SMP ×1
native	host (rtos-apis , single; smp-rtos-apis , SMP 4)	host (mutex-stress , single; smp-mutex-stress , SMP 4)	host	see §8
2xcortex-m33	QEMU	QEMU	QEMU	SMP 2
pico2-1cpu	QEMU	QEMU	QEMU	single
qemu-cortex-m0/m3/m4f/m7f	QEMU	QEMU	QEMU	single (local cortexm port's generic QEMU cores, OS_NCPU=1)
nucleo-*, raspberrypi-pico	yes	yes	yes	single (upstream); nucleo-f411re adds blinky-test

fp-switch runs where an M33 FPU does: cortexm-pico2 (qemu, hwd), 2xcortex-m33 (QEMU, SMP 2) and pico2-1cpu (QEMU, single).

On cortexm-pico2 the -qemu image of a suite does not run the board at all: QEMU has no RP2350, so the image links the generic single-core Cortex-M QEMU core in its place, and the port's builder forces NCPU to 1 for it (<cortexm>/test/CMakeLists.txt).

The validator is CMSIS-RTOS v1, which assumes one CPU. On the SMP platforms the kernel's wrapper pins every thread osThreadCreate() makes to core 0 (src/rtos/os-c-wrapper.cpp). On the Pi ports the board's harness-suite.cpp pins the main thread too; the pico2 wrapper (test/pico2/harness-suite.hpp) does not, so there the validator's main thread keeps the default affinity while every thread it creates is on core 0. On the BCM2837 (both Pi ports) its NVIC IRQ 0 is emulated with the local Mailbox 1 interrupt, and the Cortex-M DWT cycle counter it probes is answered by a RAM page that says "no cycle counter". Both exist only in the validator's image (UOS_CMSIS_OS_VALIDATOR). On native, the validator xpack's own SIGUSR1 / SIGALRM shims stand in for the NVIC.

5. AArch32 and AArch64 — Raspberry Pi Zero 2 W and Pi 3 B

The same fifteen test directories exist on all four (port, board) pairs: `<port>/test/rpi-zero-2w/` and `<port>/test/rpi3b/`. The Pi 3 B builds use the Zero 2 W's board sources with `BOARD_RPI3B`. Every test is built twice, `-qemu` and `-hwd`, except as noted.

Test	What it does	Mode
<code>smp_test0</code>	Phase-1 bring-up: PL011 UART, MMU (cacheable, shareable DRAM), the 1 ms generic-timer tick and the scheduler, with one thread printing 10 heartbeats.	SMP 4, 1 active
<code>smp_test1</code>	Semaphore ping-pong across cores: pinger on core 0, ponger on core 1, a logger on core 2 blinking the LED and printing the round count and cores.	SMP 4
<code>smp_test2</code>	Lock coherency: one worker pinned per core bumps a shared counter under a mutex; the final count must be <code>OS_NCPU × ITER</code> .	SMP 4
<code>smp_test3</code>	<code>message_queue</code> producer (core 0) and consumer (core 1) that tests primes and toggles the LED; a logger prints throughput.	SMP 4
<code>smp_test4</code>	Load balancing: CPU-bound workers with no affinity; a reporter prints each worker's per-core histogram.	SMP 4
<code>smp-mat-test</code>	Parallel block Gaussian elimination (N = 120, B = 20) across all cores with spin barriers, a thread-pool demo, and a classical LU solve as ground truth; compares and times both.	SMP 4
<code>smp-mat-sdcard-test</code>	The same solver at N = 200 with the matrices stored on and loaded from the SD card (flatfs on <code>disk.img</code> under QEMU; FatFs under <code>/tests</code> on the boot card on hardware).	SMP 4
<code>smp-num-test</code>	Five prioritised threads: UART ticker, LED blinker, an SD text writer, an FP compute writer and a reader, all sharing <code>num.txt</code> under one mutex, paced by a counting semaphore.	SMP 4
<code>smp-pipeline-test</code>	13 threads (4 producers, 4 compute workers, 2 SD writers, an auditor that checks CRC and sequence, an LED pacer, telemetry) with heavy <code>yield()</code> .	SMP 4
<code>smp-pro-cons-test</code>	Producer/consumer that uses every kernel object (thread, memory pool, message queue, both semaphores, mutex, condition variable, event flags, timer, sysclock, suspend/resume).	SMP 4
<code>sd_test</code>	SD card: under QEMU, <code>Sd::SdCard</code> + flatfs on a host-seeded <code>disk.img</code> (host ↔ device round trip); on hardware,	SMP 4

Test	What it does	Mode
	FatFs read/write under <code>/tests</code> on the boot card (never formatted).	
<code>usb_test</code>	DWC2 USB gadget file transfer: a host tool sends files over bulk OUT, stored to the SD card and hex-dumped to the UART and semihosting, with unpinned workers keeping every core busy.	SMP 4
<code>rtos-apis</code> , <code>mutex-stress</code> , <code>cmsis-os-validator</code>	The harness suites (§4).	SMP 4

Exception: `usb_test` has no `-hwd` case on the Pi 3 B, only `-qemu`. The Pi 3 B's USB ports sit behind the LAN9514 hub, so the DWC2 can only be a host.

Case counts: Zero 2 W 30 (15 × 2), Pi 3 B 29.

6. AArch32 — Luckfox Lyra B (RK3506)

Hardware only: the board declares no QEMU linker script. 19 port tests plus one harness suite, all `-hwd`, all SMP 3.

Test	What it does
<code>smp_test0</code> ... <code>smp_test4</code>	The same bring-up, ping-pong, lock-coherency, message-queue and load-balancing tests as on the Pis (§5), on three cores.
<code>smp-mat-test</code> , <code>smp-mat-sdcard-test</code> , <code>smp-num-test</code> , <code>smp-pipeline-test</code> , <code>smp-pro-cons-test</code> , <code>sd_test</code>	As on the Pis (§5), on the Lyra's SD card controller (DW-MMC).
<code>smp_test5</code>	Three pinned threads: an SD card thread on core 0 (FatFs mount, formatting FAT32 if none), an LED blinker on core 1 and a logger on core 2.
<code>smp_test6</code>	USB-to-UART logger on three cores: a DWC2 gadget task, a heartbeat logger and an LED task; driven from the host with <code>usb_test_host</code> .
<code>smp_test7</code>	USB vendor gadget receives a file, the SD task writes it as <code>received.dat</code> , then asks on the UART whether to delete it; the LED reflects the answer.
<code>smp_test_int</code>	First external interrupt: UART1 RX on the GIC (ID 99), with an ISR echo and a TX line lock shared with the ISR.
<code>smp_test_int2</code>	Interrupt-driven TX through a THRE ring, and the UART1 interrupt routed to core 1 via <code>GICD_ITARGETSR</code> .
<code>smp_test_int3</code>	PC keyboard → board: USB OTG1 as an interrupt-driven vendor gadget on core 0 (GIC ID discovered at runtime), UART1 interrupt-driven on core 1. Needs the host tool <code>kbd_forward</code> .
<code>smp_test_int4</code>	Keystroke logger with three IRQs on three cores: USB (core 0) → ring → <code>keys.log</code> on the card, written with the DW-MMC data-transfer-over IRQ (core 2); UART on core 1.
<code>smp_test_int5</code>	AMP doorbell: the RK3506's Cortex-M0 fires Mailbox interrupts every ~40 ms to A7 cores 1 and 2, which report to core 0 over a shared ring.
<code>mutex-stress-test</code>	The harness suite, built by the platform itself (not a port test directory).

Every Lyra test ends with a `RESULT:` line and a semihosting exit. The four USB tests (`smp_test6`, `smp_test7`, `smp_test_int3`, `smp_test_int4`) wait up to 120 s for the host to configure the gadget, then fail; they are built `UOS_CONSOLE_UART_ONLY`, so their console does not mirror every line through semihosting (each such trap halts the core, and during USB traffic that lost the WCH-Link's debug link). Their host tools are C/C++ sources in the test's own `host/` directory: `usb_test_host` for `smp_test6` / `smp_test7`, `kbd_forward` for `smp_test_int3` / `smp_test_int4`.

7. Cortex-M

7.1 RP2350 boards

Three boards on the same silicon (2× Cortex-M33): `pico2`, `pico2-pizero` and `pico2-rp2350b-psram`. On the single-core tests `OS_NCPU` is 1 (`board_test_ncpu`); the "bare" tests link no kernel (`BOARD_TEST_NO_KERNEL`).

Test	What it does	Mode	pico2	pizero	rp2350b-psram
<code>exc-test</code>	Bare exception-entry probe: SysTick with MSP, then PendSV with PSP.	bare	hwd	hwd	hwd
<code>smp-test0</code>	Bare boot chain and dual-core launch: C++ constructors, 150 MHz clocks, UART; core 1 blinks the LED.	bare, 2 cores	hwd	hwd	hwd
<code>smp-test1</code>	Single-core RTOS bring-up: three threads, sysclock, mutex, counting semaphore, LED heartbeat.	single	qemu, hwd	hwd	qemu, hwd
<code>sc-test-ko</code>	Single-core kernel-object test: ten self-checking sub-tests (mutex, semaphores, queues, event flags, timers, ...), controller and helper on core 0.	single	qemu, hwd	hwd	qemu, hwd
<code>smp-test-ko</code>	The SMP twin of <code>sc-test-ko</code> : controller pinned to core 0, helper to core 1, so every hand-off is cross-core.	SMP 2	hwd	hwd	hwd
<code>smp-test2</code>	Dual-core SMP: per-core PendSV and SysTick, affinity (one worker per core), SIO spinlock-backed kernel lock.	SMP 2	hwd	hwd	hwd
<code>smp-test3</code>	Cross-core IPI through the SIO FIFO: semaphore ping-pong with the one-way wake latency measured on TIMER0.	SMP 2	hwd	hwd	hwd

Test	What it does	Mode	pico2	pizero	rp2350b-psram
<code>smp-test4</code>	Kernel objects under dual-core contention: seven round-trip sub-tests between a controller (core 0) and a worker (core 1).	SMP 2	hwd	hwd	hwd
<code>smp-test5</code>	First peripheral interrupt: UART0 TX IRQ serviced on core 1, draining a ring filled by core 0.	SMP 2	hwd	hwd	hwd
<code>smp-mat-test</code>	Parallel block solver ($N = 100$, $B = 50$) on two cores. On <code>pico2-rp2350b-psram</code> its data lives in PSRAM.	SMP 2	hwd	hwd	hwd
<code>smp-test-usb-cdc-acm</code>	USB CDC-ACM virtual serial port: echoes what the host types, mirrored on the UART.	SMP 2	hwd	hwd	—
<code>smp-test-usb-hid</code>	USB HID keyboard forwarding; keystrokes printed on the UART.	SMP 2	hwd	hwd	—
<code>psram-exec</code>	Bare SRAM supervisor that copies two apps into PSRAM and switches between them on a key press.	bare	—	hwd	—
<code>psram-mat-test-250</code>	<code>smp-mat-test</code> with PSRAM brought up by the app itself, at 250 MHz, from an SRAM-boot linker script.	SMP 2	—	hwd	—
<code>smp-test-nested</code>	Nested interrupts on both cores (SysTick, UART0,	SMP 2	—	—	hwd

Test	What it does	Mode	pico2	pizero	rp2350b-psram
	TIMER0 IRQ 2/3 at different priorities) with thread migration and a lock-free trace. Passes once both workers have run 20 beats and all three nesting interrupts have fired.				
<code>smp-test-nested-clock</code>	<code>smp-test-nested</code> plus a clock tree driven from a potentiometer (ADC) and the on-die temperature sensor; data in PSRAM, 250 MHz enabled. Interactive: waits for <code>y</code> on the UART (a pot position other than the current clock reboots into it first), then the same 20-beat verdict.	SMP 2	—	—	hwd
<code>smp-test-nested-clock_200</code>	The same, without <code>PICO2_ENABLE_250MHZ</code> (the clock tree at its default).	SMP 2	—	—	hwd
<code>smp-test-nested-clock_250</code>	The same, with 250 MHz enabled.	SMP 2	—	—	hwd
<code>rtos-apis</code> , <code>mutex-stress</code> , <code>cmsis-os-validator</code> , <code>fp-switch</code>	The harness suites (§4).	SMP 2 (hwd); single (qemu)	qemu, hwd	—	—
<code>cmsis-os-validator-ram</code> , <code>mutex-stress-ram</code> , <code>rtos-apis-ram</code>	RAM-resident harness suites executing entirely from internal SRAM (no flash writes).	single	hwd	—	—
<code>smp-mat-test-ram</code>	Parallel block solver executing	SMP 2	hwd	—	—

Test	What it does	Mode	pico2	pizero	rp2350b-psram
	entirely from internal SRAM (no flash writes).				

Case counts: `pico2` 26, `pico2-pizero` 14, `pico2-rp2350b-psram` 16.

7.2 STM32F4 boards (one CPU)

`cortexm-nucleof411` and `cortexm-weactf411` run both on hardware and on QEMU (`-M netduinoplus2 -cpu cortex-m4`). `cortexm-weactf412` is hardware-only. The harness suites build the SMP scheduler at `OS_NCPU = 1`; the board's own tests use the kernel's non-SMP branch.

Test	What it does	Mode	nucleof411	weactf411	weactf412
<code>mos-test1</code>	Single-CPU acceptance: producers and consumers over a bounded buffer with plain and recursive mutexes, counting semaphores, sysclock and an LED thread.	single	qemu, hwd	qemu, hwd	hwd
<code>spi-pipeline</code>	SPI filesystem pipeline (ported from a Rust example): five threads, eight semaphores, Base64 encode to partition 1, decode and CRC check to partition 2.	single	—	qemu, hwd	—
<code>uart-test1</code>	USART1 echo: RX interrupt → bounded buffer → TX via DMA2, with an LED thread.	single	—	—	hwd
<code>rtos-apis</code> , <code>mutex-stress</code> , <code>cmsis-os-validator</code>	The harness suites (§4).	SMP ×1	qemu, hwd	qemu, hwd	hwd

QEMU execution on `netduinoplus2`: - `nucleof411`: 4/4 PASS (`mos-test1`, `rtos-apis`, `mutex-stress`, `cmsis-os-validator`). - `weactf411`: 5/5 PASS (`mos-test1`, `spi-pipeline`, `rtos-apis`, `mutex-stress`, `cmsis-os-validator`). - `weactf412` **hardware-only rationale:** 1. *SRAM limit:* STM32F412RE has 256 KB of SRAM, setting `_estack` at `0x20040000`. QEMU's `netduinoplus2`

machine allocates only 128 KB of SRAM (`0x20000000` – `0x20020000`), causing QEMU to fault on startup.

2. *Peripheral support*: `uart-test1` uses DMA2 Stream 7 (`DMA2_Stream7`) for USART1 TX, which is not implemented in the QEMU Netduino Plus 2 model.

Runtime termination fix (`hw_result.hpp`): In `micro-os-plus-iii-cortexm/test/boards/shared/hw_result.hpp`, tests terminate using `std::_Exit(0)` instead of `std::exit(0)`. Normal `std::exit()` invokes `atexit` handlers and static object destructors while RTOS threads are still alive, causing `assert(empty())` assertion failures in `double_list::~double_list()`. `std::_Exit(0)` performs a clean termination and semihosting exit immediately.

8. POSIX — native host

`micro-os-plus-iii-posix-arch/test/native/`. Host threads stand in for CPUs: `NCPU` defaults to 4 (`-DNCPU=1` builds everything single-core). All cases are `-host`.

Test	What it does	Mode
<code>smp_test0</code> ... <code>smp_test4</code>	The Pi tests of the same names (§5), on host threads.	SMP 4 (<code>smp_test0</code> : 1 active)
<code>smp-mat-test</code> , <code>smp-num-test</code> , <code>smp-pipeline-test</code> , <code>smp-pro-cons-test</code>	As on the Pis; the SD card is flatfs over a host file.	SMP 4
<code>mutex-stress</code>	The upstream suite, unchanged.	single
<code>rtos-apis</code>	The upstream suite, with the POSIX I/O layer.	single
<code>smp-mutex-stress</code>	The SMP leg of <code>mutex-stress</code> , with <code>smp_test2</code> 's start-up.	SMP 4
<code>smp-rtos-apis</code>	The whole <code>rtos-apis</code> sequence once per core, from a driver pinned to that core, with the API threads unpinned; fails unless every core did the work.	SMP 4
<code>flatfs-test</code>	flatfs on a host-file card: append 64 KiB to a file created empty, then check it, a file beside it and a fresh mount.	SMP 4
<code>mutex-ceiling-test</code>	A thread above a robust protect mutex's ceiling gets <code>EINVAL</code> ; after it is destroyed, lock/unlock at the ceiling must still work (no <code>EOWNERDEAD</code>).	SMP 4
<code>cmsis-os-validator</code>	The validator (§4), registered by the harness platform itself.	SMP 4, validator on core 0

16 cases.

9. How to run

From repository root `micro-os-plus-iii` (or `tests/`), for a configuration `C` such as `aarch64-rpi3b-cmake-gcc-debug`:

```
xpm run install --config C && xpm run prepare --config C && xpm run build --config C
xpm run test --config C                        # every qemu/host case (-LE hwd)
xpm run test-<app>-<variant> --config C        # one case, builds first
PATH="$PWD/build/C/xpacks/.bin:$PATH" ctest --test-dir build/C -N  # list the cases
```

For a `-hwd` case, power-cycle the board first and run exactly one case. On the Pis choose the probe with `PROBE=jlink|olimex`:

```
cd build/C && PROBE=olimex PATH="$PWD/xpacks/.bin:$PATH" ctest -V -R <platform>-<app>-hwd
```